

Sarah Alnegheimish

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Research Interests: Machine Learning Frameworks, Deep Learning, Time Series, Natural Language Processing.

EDUCATION

2022 - 2025	Massachusetts Institute of Technology Ph.D. in Electrical Engineering and Computer Science (EECS) <i>Supervised by Dr. Kalyan Veeramachaneni</i>	GPA: 5.0/5.0
2019 - 2022	Massachusetts Institute of Technology S.M. in Electrical Engineering and Computer Science (EECS) S.M. in Computational Science and Engineering (CSE) <i>Supervised by Dr. Kalyan Veeramachaneni</i>	GPA: 5.0/5.0
2013 - 2017	King Saud University B.Sc. in Computer Science <i>First Class Honors</i>	GPA: 5.0/5.0

EXPERIENCE

Research Assistant | DAI Lab, MIT *Cambridge, MA*
Machine Learning Frameworks for Time Series Tasks (PhD Thesis) *Sep 2022 - present*

- Leading *Sintel* ecosystem for time series modeling and analysis.
- Designed and implemented universal representations of *primitives* and *pipelines* for time series tasks including: classification, forecasting, and anomaly detection.
- Conducted real-world case studies featuring two predictive maintenance studies on wind turbine data and two studies on anomaly detection for satellite data and electric vehicle data.
- Collaborated in the development of *Zephyr*, a data-centric framework for predictive maintenance of wind turbine energy data, with generalized abstractions to define various prediction problems.

Unsupervised Time Series Anomaly Detection (Master Thesis) *Sep 2019 - May 2022*

- Developed *Orion*, an open-source library for unsupervised time series anomaly detection ([github stars: 1,000](#)).
- Created a thorough benchmark suite for continuous quality and computational evaluations.
- Designed metrics for time-based anomaly scoring that prioritize the anomalous class.
- Collaborated with students on a number of anomaly detection pipelines including: an autoencoder with regression model (AER) and a generative adversarial model (TadGAN) for reconstruction-based anomaly detection.

Gender Bias in Large Language Models *Sep 2021 - Feb 2022*

- Created a new dataset to examine model components that elicit biased output.
- Proposed mitigation strategies to reduce the discrimination severity of the model output.

Intuitive Physics for Game Play Bots *Sep 2019 - May 2020*

- Built a probabilistic program to infer actions performed by an agent in Newtonian settings.
- Complemented the program with a model-free approach for efficient sampling.

Applied Scientist Intern | Amazon *Bellevue, WA*
Multivariate Time Series Anomaly Detection *May 2024 - Aug 2024*

- Developed a multivariate time series model to detect anomalies from multiple systems pertaining to a single asset.
- Packaged the model in a domain-specific framework for seamless end-to-end usage and benchmarking.

Software Engineer Intern | DataCebo *Boston, MA*
Modeling Generative Time Series *Jun 2021 - Aug 2021*

- Worked on improving the modularization of the PARModel which is a generative model for synthetic time series.
- Worked closely with the SDV team on reversible data transforms and conditional sampling.

Research Specialist | Center for Complex Systems at KACST and MIT *Riyadh, Saudi Arabia*
Predictive Framework for Healthcare Data *Dec 2017 - Aug 2019*

Collaborated with Dr. Kalyan Veeramachaneni

- Developed *Cardea*, an automated machine learning library to solve health related prediction problems.
- Integrated HL7's *Fast Healthcare Interoperability Resources* standard as a representation for data.

Occupation-Skill Space

Dec 2017 - Aug 2019

Collaborated with Professor Iyad Rahwan

- Constructed an occupation network by discovering job-job relationships through their underlying skills.
- Utilized the job space as a function to predict individual career trajectories.
- Analyzed the impact of automation on job mobility and the education hierarchy.

Data Analyst | Mozn

Riyadh, Saudi Arabia

Fraud Detection and Optical Character Recognition

May 2017 - Dec 2017

- Worked closely with a team of experts to develop a system that detects fraudulent shipments.
- Curated a dataset of Arabic characters and trained a convolutional model for automated character recognition.

PUBLICATIONS

- **Alnegheimish, S.**, Nguyen, L., Berti-Equille, L. and Veeramachaneni, K., *Can Large Language Models be Anomaly Detectors for Time Series?*. IEEE Conference on Data Science and Advanced Analytics (DSAA), 2024. (accepted).
- **Alnegheimish, S.**, Liu, D., Sala, C., Berti-Equille, L. and Veeramachaneni, K., *Sintel: A Machine Learning Framework to Extract Insights from Signals*. ACM SIGMOD International Conference on Management of Data, 2022.
- **Alnegheimish, S.***, Guo, A.* , Sun, Y.* , *Using Natural Sentence Prompts for Understanding Biases in Language Models*. Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2022.
- Alhasoun, F. and **Alnegheimish, S.**, *Probabilistic Programming Bots in Intuitive Physics Game Play*. 35th AAAI Conference on Artificial Intelligence, 2021.
- Geiger, A., Liu, D., **Alnegheimish, S.**, Cuesta-Infante, A., Veeramachaneni, K., *TadGAN: Time Series Anomaly Detection Using Generative Adversarial Networks*. IEEE Conference on Big Data, 2020. (citations: 311).
- **Alnegheimish, S.**, Alrashed, N., Aleissa, F., Althobaiti, S., Liu, D., Alsaleh, M. and Veeramachaneni, K., *Cardea: An Open Automated Machine Learning Framework for Electronic Health Records*. IEEE Conference on Data Science and Advanced Analytics (DSAA), 2020.

Preprints

- **Alnegheimish, S.**, Berti-Equille, L. and Veeramachaneni, K., *Making the End-User a Priority in Benchmarking: OrionBench for Unsupervised Time Series Anomaly Detection*.

HONORS & AWARDS

- MIT's EECS Graduate Alumni Fellowship, 2022-2023.
- Graduate Scholarship, King Abdulaziz City for Science and Technology (KACST), 2019 - 2022.
- Google's CS Research Mentorship Program (CSRMP), 2020 - 2021.
- Graduate Fellow, MiSK, 2019 - 2022.
- 2nd place at AEC annual best graduation project, 2017.
- Best graduation project in the college of Computer and Information Sciences, King Saud University, 2017.
- Best poster at the 9th Undergraduate Research Conference, Zayed University, 2017.
- Dean's list for outstanding students at King Saud University, 2015 and 2016.

SKILLS

Programming Languages	Python, Tensorflow, Pytorch, MATLAB, Java, R, C/C++, JavaScript, HTML, SQL English (native), Arabic (native)
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MEDIA

MIT News

- MIT researchers use large language models to flag problems in complex systems. August 14, 2024
- One-stop machine learning platform turns health care data into insights. April 9, 2021
- Method finds hidden warning signals in measurements collected over time. December 17, 2020